Akoizo Original Practice - Form 2

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which is a controlled variable rather than a control treatment?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. A baseline reference level

B. An untreated comparison

C. Keeping light constant

D. A no-added-fertilizer group

Answer: _____

2. (2 points) What conclusion is supported by these trials?

Unit: A clear Division B investigation report.
Worked-case data: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.
Separate investigation observations:
25 cm^2 towel: Absorbed masses: 2, 3, 4 g.
50 cm^2 towel: Absorbed masses: 5, 6, 7 g.
Controls: Same towel brand, soak time, and draining method.

A. The larger towel absorbed less water

B. Soak time caused the change because it was increased

C. Every individual measurement doubled

D. Doubling area increased mean absorbed mass from 3 g to 6 g

Answer: _____

3. (2 points) Standard deviation has…

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. The independent-variable units necessarily

B. The response's units

C. Squared response units

D. No units always

Answer: _____

4. (2 points) A failed hypothesis means the experiment is invalid?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn "study motion" into a testable question.

A. Only if data vary

B. Only if the graph is flat

C. No

D. Yes

Answer: _____

5. (2 points) A conclusion should answer…

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

A. The original measured question

B. Every possible mechanism

C. A preferred hypothesis regardless of data

D. A different untested outcome

Answer: _____

6. (4 points) Improve "release and time it."

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

Answer: _____
_____
_____

7. (4 points) Name three dependent reporting tasks.

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
_____
_____

8. (4 points) Give a variance mechanism for timing.

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
_____
_____

9. (4 points) Why not invent qualitative notes after analysis?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

Answer: _____
_____
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10. (4 points) What should a changed procedure record?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
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11. (4 points) What limits extrapolation?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
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_____

12. (6 points) Solve this case and explain your reasoning: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
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13. (2 points) Define “Trial” in the context of Experimental Design.

Answer: _____
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14. (2 points) Define “Claim” in the context of Experimental Design.

Answer: _____
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15. (2 points) Define “Controlled variable” in the context of Experimental Design.

Answer: _____
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16. (2 points) Define “Coherence” in the context of Experimental Design.

Answer: _____
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17. (2 points) Define “Reasoning” in the context of Experimental Design.

Answer: _____
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18. (2 points) Define “Precision” in the context of Experimental Design.

Answer: _____
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